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MTH 424 — Past Questions & Materials
by Renance

Course: MTH 424

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 2

Levels: 400

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Paper 1 of 2 — MTH 424 System Theory Second Semester 2020 2021

Level 400 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS

SECOND SEMESTER EXAMINATION 2020/2021

Course Code/Title: MTH 424 –System Theory

Instructions: Answer any 4 questions. **Time Allowed:** 2½ Hours. **NOTE:** 30 Marks for C.A

Question 1

- Define Frechet derivative in a normed linear spaces. Find the Frechet derivative of $g(x) = xy + x^2$ at the point x with increment h , where $x \in (x, y)$, $h = (h_1, h_2)$ (4 ½ marks)
- Let $\dot{x} = F(x, t)$ be a system of differential equation. When is the system said to be (i) autonomous (ii) non-autonomous (iii) stable (iv) unstable. Define critical point denoted by x_c . When is the critical point x_c said to be (i) stable (ii) asymptotically stable (iii) quasi-asymptotically (7 marks)
- Express the following differential equations as a system of first order differential equations in matrix form: (i) $\ddot{y} + a\dot{y} = 0$, $y(0) = y_0$, $\dot{y}(0) = y_1$ (ii) $\dot{y} + y = 0$, $y(0) = y_0$, $\dot{y}(0) = y_1$ (6 marks)

Question 2

- Define a controllable space (3 ½ marks)
- Show that the controllability space function of the system $\dot{y} = Ay(t) + Bx(t)$, $y(0) = y_0$ is a linear subspace of R^n , $n \geq 1$. (8marks)
- Define absolutely continuous function which satisfies $\dot{y} = Ay(t) + Bx(t)$, $y(0) = y_0$. State when this system is said to be controllable (6 marks)

Question 3

- Define the norm function $(\|\cdot\|)$. Prove that $\|x\| = (|x_1|^2 + |x_2|^2 + \dots + |x_n|^2)^{\frac{1}{2}}$ is a norm (6marks)
- What do you understand by the term *normed linear space*? Give an example of this space (6marks)
- Define the following terms in a normed linear space (i) Compactness (ii) Relative compactness (iii) completeness (iv) Denseness (5 ½ marks)

Question 4

- Let $\dot{x} = f(t, x)$ be a system of differential equations where f is continuous on the domain D and the solution $x(t)$ is defined for all $t > \tau$. Then state the Lyapunov stability conditions (6 marks)
- If the conditions above are satisfied, then when is the solution $x(t, t_0, x^0)$ said to be asymptotically conditionally stable in the sense of Lyapunov? (3 ½ marks)
- Investigate the stability of the following systems in the sense of Lyapunov: (i) $\dot{x} = 0$ (ii) $\dot{x} = x^2$ (iii) $\dot{x} = 1 - x^2$ (8 marks)

Question 5

- Define (i) the Jacobian matrix of a continuous vector valued function defined on a subset S of R^n (ii) The Hessian matrix of a continuous vector valued function defined on a subset S of R^n (4marks)
- Consider the function $F = (f, g, h, u)$, where $f(x, y, z) = x^2 + y^2 + z^2$, $g(x, y, z) = x \sin y + \cos z$, $h(x, y, z) = (x + y) \sin z$ and $u(x, y, z) = (y + z) + \cos xy$. Then find the Jacobian matrix (6 marks)
- Find the Hessian matrix of the function above. When is the Hessian matrix said to (i) positive definite (ii) positive semi definite (iii) negative definite (iv) negative semi definite (7 ½ marks)

Question 6

- Define (i) Symmetric matrix (ii) Positive matrix (iii) non-negative matrix (3 marks)
- Let $\dot{x} = Ax$, $x(0) = x$, $x \in R^n$. Consider $\dot{v} = -\langle x, Cx \rangle$, show that $PA + A^T P + C = 0$ (7 marks)
- Given that $\dot{x} = \begin{pmatrix} -\frac{1}{2} & 1 \\ 0 & -1 \end{pmatrix} x$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. Then find P and C . What conditions would be placed on P and C for the origin of the given linear system to be asymptotically stable? (7 ½ marks)

Paper 2 of 2 — MTH 424 System Theory Second Semester 2024 2025

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